

ASS:-14.3

NAME:-B.PRATHAMKOUSHIK

ROLLNO:-2403A52182

Task Description #1 – AI-generated HTML Page

Task: Ask AI to generate a simple HTML homepage for a "Student Info

Portal" with a header, navigation menu, and footer.

Expected Output:

- HTML code with <header>, <nav>, <footer>.
- Clean indentation, proper tags, and comments.

PROMPT:

Task: Ask AI to generate a simple HTML homepage for a "Student Info Portal" with a header, navigation menu, and footer

CODE:

```
Commands + Code + Text ▶ Run all ▼

[4] ✓ Os from IPython.display import HTML

html_content = """
<!DOCTYPE html>
<html>
<head>
  <title>Student Info Portal</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
    }
    header {
      background-color: #f2f2f2;
      padding: 10px;
      text-align: center;
    }
    nav {
      background-color: #e0e0e0;
      padding: 10px;
      text-align: center;
    }
    nav a {
      margin: 0 15px;
      text-decoration: none;
      color: #333;
    }
    footer {
      background-color: #f2f2f2;
padding: 10px;
text-align: center;
position: fixed;
bottom: 0;
width: 100%;
    }
  </style>
</head>
<body>

  <header>
    <h1>Student Info Portal</h1>
  </header>

  <nav>
    <a href="#">Home</a>
    <a href="#">Profile</a>
    <a href="#">Courses</a>
    <a href="#">Grades</a>
  </nav>

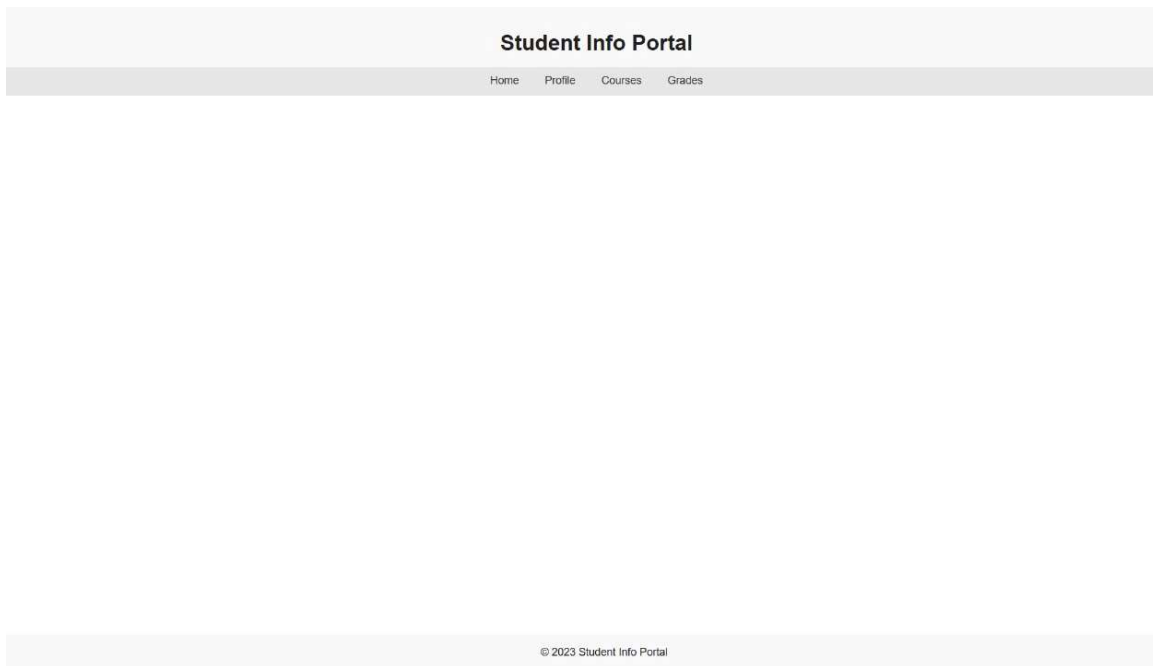
  <main>
    <!-- Content goes here -->
  </main>

  <footer>
    <p>&copy; 2023 Student Info Portal</p>
  </footer>

</body>
</html>

```

OUTPUT:



Task Description #2 – CSS Styling

Task:

Use AI to add CSS styling to Task #1 homepage for:

- Responsive navigation bar.
- Centered content section.
- Footer with light gray background.

Expected Output:

- HTML + CSS combined.
- AI explains how CSS classes apply.

Expected Output: AI refactors with with open() and try-except:

PROMPT:

Task: Use AI to add CSS styling to Task #1 homepage for: • Responsive navigation bar. • Centered content section. • Footer with light gray background. Expected Output: • HTML + CSS combined. • AI explains how CSS classes apply. Expected Output: AI refactors with with open() and try-except:

CODE:

```
from IPython.display import HTML

html_content = """
<!DOCTYPE html>
<html>
<head>
<title>Student Info Portal</title>
<style>
    body {
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        line-height: 1.6;
    }
    header {
        background-color: #f2f2f2;
        padding: 10px;
        text-align: center;
    }
    nav {
        background-color: #e0e0e0;
        padding: 10px;
        text-align: center;
    }
    nav a {
        margin: 0 15px;
        text-decoration: none;
        color: #333;
        display: inline-block; /* Added for better spacing */
    }
}

/* Responsive Navigation */
@media (max-width: 600px) {
    nav a {
        display: block;
        margin: 5px 0;
    }
}

.container {
    max-width: 960px; /* Maximum width of content */
    margin: 20px auto; /* Center the container and add top/bottom margin */
    padding: 0 20px; /* Add some padding on the sides */
}

footer {
    background-color: #f2f2f2; /* Light gray background */
    padding: 10px;
    text-align: center;
    position: fixed;
    bottom: 0;
    width: 100%;
}

</style>
</head>
<body>

<header>
<h1>Student Info Portal</h1>
</header>

<nav>
<a href="#">Home</a>

```

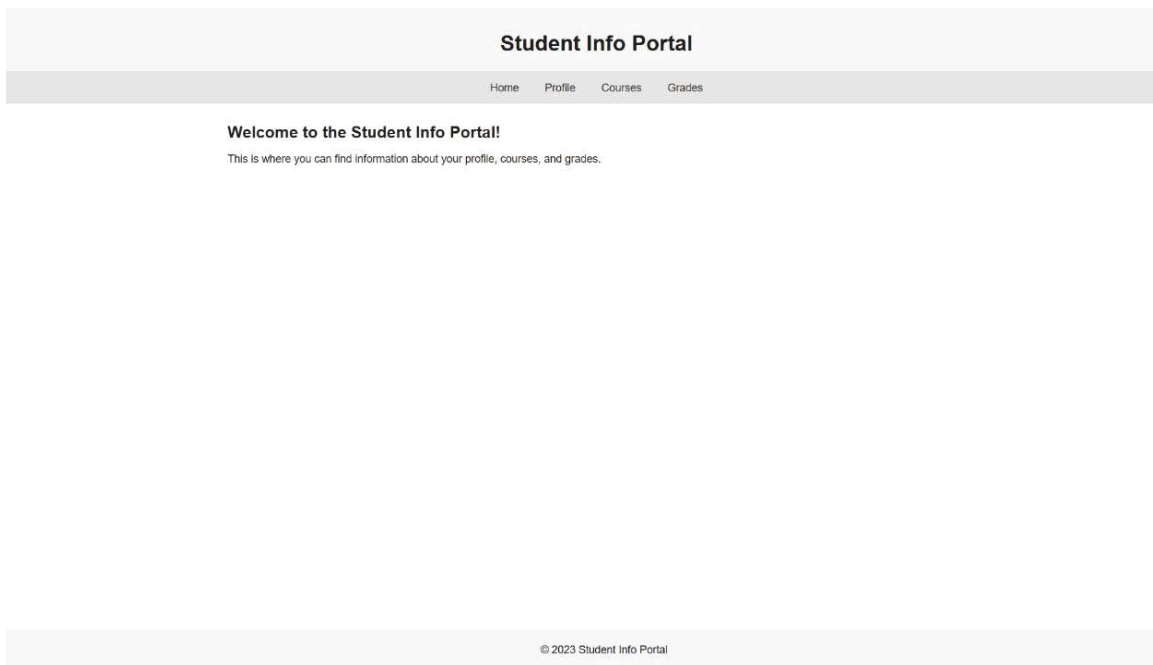
```
<a href="#">Grades</a>
</nav>

<div class="container">
  <main>
    <h2>Welcome to the Student Info Portal</h2>
    <p>This is where you can find information about your profile, courses, and grades.</p>
    <!-- More content can be added here -->
  </main>
</div>

<footer>
  <p>&copy; 2023 Student Info Portal</p>
</footer>

</body>
</html>
<!--
disolv(HTML(html content))
```

OUTPUT:



Task Description #3 – JavaScript Interactivity

Task: Prompt AI to generate a JS script that validates a simple login form (non-empty username/password).

Expected Output:

Working on submit JS validation.

Clear error messages if inputs are empty.

PROMPT:

Task Description #3 – JavaScript Interactivity Task: Prompt AI to generate a JS script that validates a simple login form (non-empty username/password). Expected Output: Working on submit JS validation. Clear error messages if inputs are empty. give output

CODE:

```
# Generate HTML form
html_form = """
<!DOCTYPE html>
<html>
<head>
<title>Login Form</title>
<style>
  body {
    font-family: sans-serif;
  }
  .login-container {
    width: 300px;
    margin: 50px auto;
    padding: 20px;
    border: 1px solid #ccc;
    border-radius: 5px;
    background-color: #f9f9f9;
  }
  .login-container h2 {
    text-align: center;
    color: #333;
  }
  .login-container label {
    display: block;
    margin-bottom: 5px;
    color: #555;
  }
  .login-container input[type="text"],
  .login-container input[type="password"] {
    width: 100%;
    padding: 10px;
    margin-bottom: 10px;
    border: 1px solid #ccc;
    border-radius: 3px;
  }
  .login-container button {
    width: 100%;
    padding: 10px;
    background-color: #5cb85c;
    color: white;
    border: none;
    border-radius: 3px;
    cursor: pointer;
  }
  .login-container button:hover {
    background-color: #4cae4c;
  }
  #error-message {
    color: red;
    margin-top: 10px;
    text-align: center;
  }
</style>
</head>
<body>

<div class="login-container">
  <h2>Login Form</h2>

  <form id="loginForm" onsubmit="return validateForm()">
    <div>
```

```

    <br>
    <div>
        <label for="password">Password:</label><br>
        <input type="password" id="password" name="password">
    </div>
    <br>
    <button type="submit">Login</button>
    <div id="error-message"></div>
</form>
</div>

</body>
</html>
"""

# Generate JavaScript
js_validation_code = """
<script>
function validateForm() {
    let username = document.getElementById("username").value;
    let password = document.getElementById("password").value;
    let errorMessage = document.getElementById("error-message");

```

```

    <br>
    <div>
        <label for="password">Password:</label><br>
        <input type="password" id="password" name="password">
    </div>
    <br>
    <button type="submit">Login</button>
    <div id="error-message"></div>
</form>
</div>

</body>
</html>
"""

# Generate JavaScript
js_validation_code = """
<script>
function validateForm() {
    let username = document.getElementById("username").value;
    let password = document.getElementById("password").value;
    let errorMessage = document.getElementById("error-message");

    if (username == "") {
        errorMessage.textContent = "Username must be filled out";
        return false;
    }

    if (password == "") {
        errorMessage.textContent = "Password must be filled out";
        return false;
    }

```

```

    }

    errorMessage.textContent = ""; // Clear error message on success
    return true;
}
</script>
"""

# Combine HTML and JavaScript
combined_html = html_form.replace('</body>', f'{js_validation_code}\n</body>')

# Save and display
file_path = 'login_form.html'

try:
    with open(file_path, 'w') as f:
        f.write(combined_html)
    print(f'Successfully wrote {file_path}')
except IOError as e:
    print(f'Error writing to file: {e}')

try:
    with open(file_path, 'r') as f:
        saved_content = f.read()
    print(f'\nContent of {file_path}:')
    print(saved_content)
except IOError as e:
    print(f'Error reading from file: {e}')

```

➡ Successfully wrote login_form.html

Content of login form.html:

```

<!DOCTYPE html>
<html>
<head>
<title>Login Form</title>
<style>
    body {
        font-family: sans-serif;
    }
    .login-container {
        width: 300px;
        margin: 50px auto;
        padding: 20px;
        border: 1px solid #ccc;
        border-radius: 5px;
        background-color: #f9f9f9;
    }
    .login-container h2 {
        text-align: center;
        color: #333;
    }
    .login-container label {
        display: block;
        margin-bottom: 5px;
        color: #555;
    }
    .login-container input[type="text"],
    .login-container input[type="password"] {
        width: 100%;
        padding: 10px;
        margin-bottom: 10px;
        border: 1px solid #ccc;
        border-radius: 3px;
    }
    .login-container button {

```



```
width: 100%;
padding: 10px;
background-color: #5cb85c;
color: white;
border: none;
border-radius: 3px;
cursor: pointer;
}
.login-container button:hover {
  background-color: #4cae4c;
}
#error-message {
  color: red;
  margin-top: 10px;
  text-align: center;
}
</style>
</head>
<body>

<div class="login-container">
  <h2>Login Form</h2>

  <form id="loginForm" onsubmit="return validateForm()">
    <div>
      <label for="username">Username:</label><br>
      <input type="text" id="username" name="username">
    </div>
    <br>
    <div>
      <label for="password">Password:</label><br>
      <input type="password" id="password" name="password">
    </div>
    <hr>

    <div>
      <label for="password">Password:</label><br>
      <input type="password" id="password" name="password">
    </div>
    <br>
    <button type="submit">Login</button>
    <div id="error-message"></div>
  </form>
</div>
```

OUTPUT:

Login Form

Username:

Password:

Login